Untitled

2025-10-03

```
library(foreign)
library(e1071)
```

```
## Warning: package 'e1071' was built under R version 4.4.3
```

```
library(mclust)
```

```
## Warning: package 'mclust' was built under R version 4.4.3
```

```
## Package 'mclust' version 6.1.1
## Type 'citation("mclust")' for citing this R package in publications.
```

```
library(BayesCVI)
```

```
## Warning: package 'BayesCVI' was built under R version 4.4.3
```

```
library(UniversalCVI)
```

```
## Warning: package 'UniversalCVI' was built under R version 4.4.3
```

```
library(dbscan)
```

```
## Warning: package 'dbscan' was built under R version 4.4.3
```

```
##
## Attaching package: 'dbscan'
```

```
## The following object is masked from 'package:stats':
##
##     as.dendrogram
```

data 1 (10 class)

cluster บังคับเลือก 10 กลุ่ม

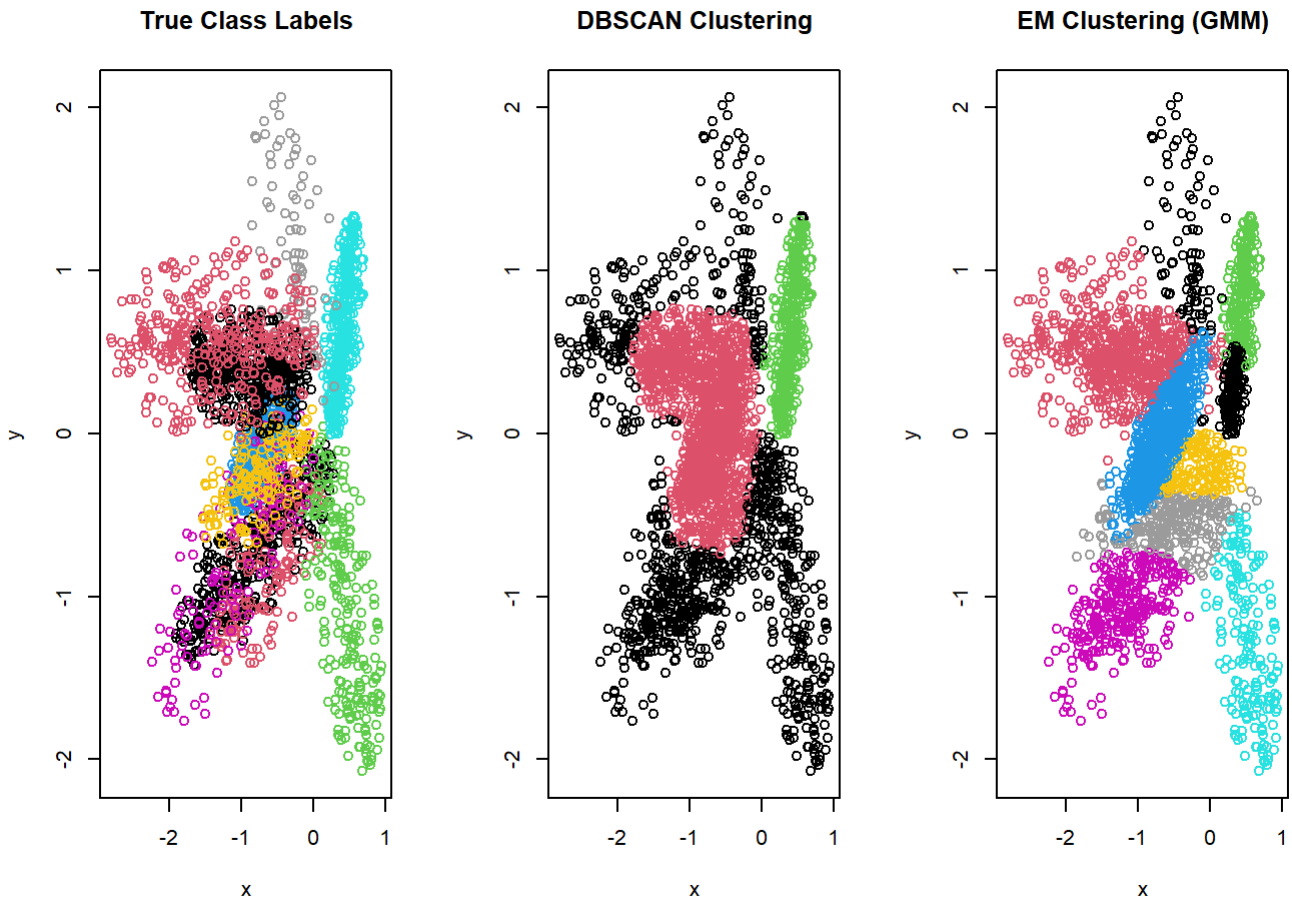
ได้ผลว่า EM สามารถแบ่งกลุ่มข้อมูลได้ดีที่สุด ด้วย Accuracy = 54.7%

```r
par(mfrow = c(1,3))


# True Labels
plot(dat1[,1:2], col = dat1$class, main = "True Class Labels")

# DBSCAN
plot(dat1[,1:2], col = dat1_dbc$cluster + 1, main = "DBSCAN Clustering")

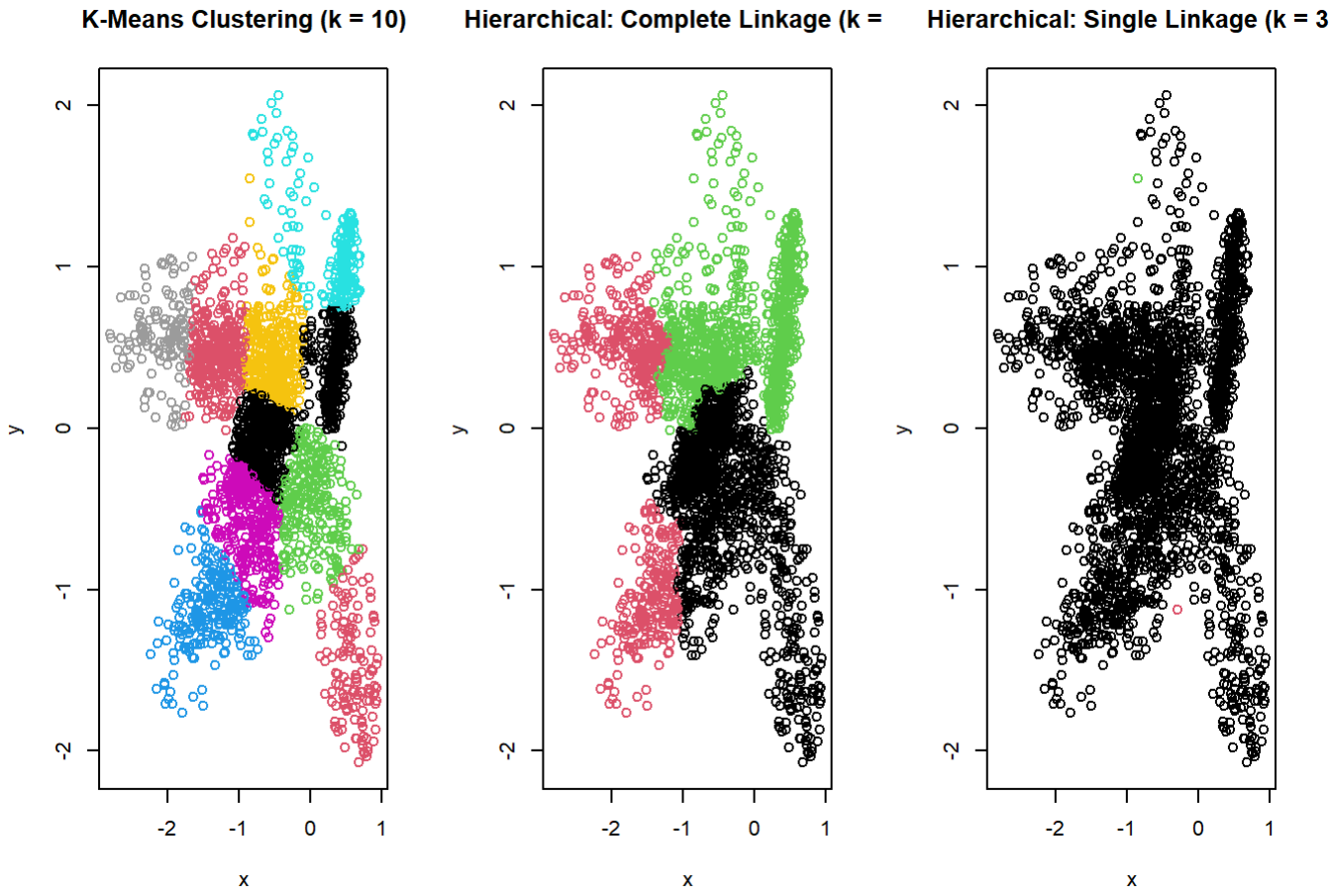
# EM (GMM)
plot(dat1[,1:2], col = dat1_EM$classification, main = "EM Clustering (GMM)")
```




```r
# K-Means
plot(dat1[,1:2], col = dat1_km$cluster, main = "K-Means Clustering (k = 10)")

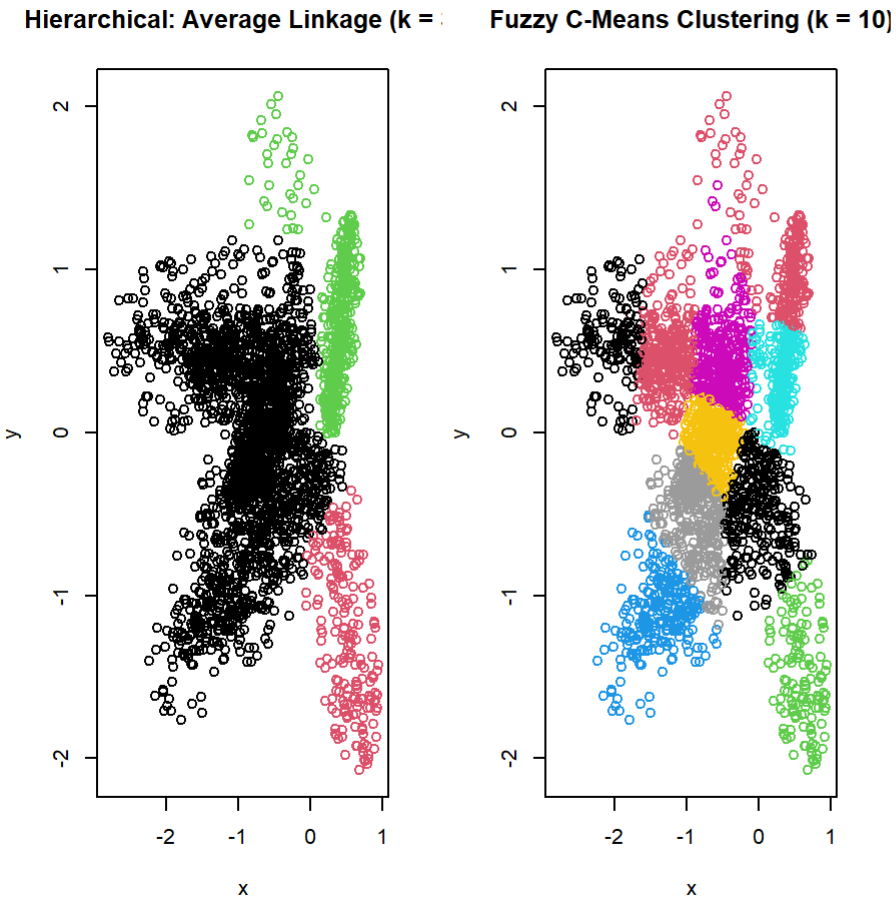
# Hierarchical: Complete Linkage
plot(dat1[,1:2], col = cutree(dat1_com, 3), main = "Hierarchical: Complete Linkage (k = 3)")

# Hierarchical: Single Linkage
plot(dat1[,1:2], col = cutree(dat1_avg, 3), main = "Hierarchical: Single Linkage (k = 3)")
```




```
# Hierarchical: Average Linkage
plot(dat1[,1:2], col = cutree(dat1_sin, 3), main = "Hierarchical: Average Linkage (k = 3)")

# Fuzzy C-Means
plot(dat1[,1:2], col = dat1_fz$cluster, main = "Fuzzy C-Means Clustering (k = 10)")
```



---

data 2 (4 class + “noise” class)

cluster บังคับเลือก 4 กลุ่ม

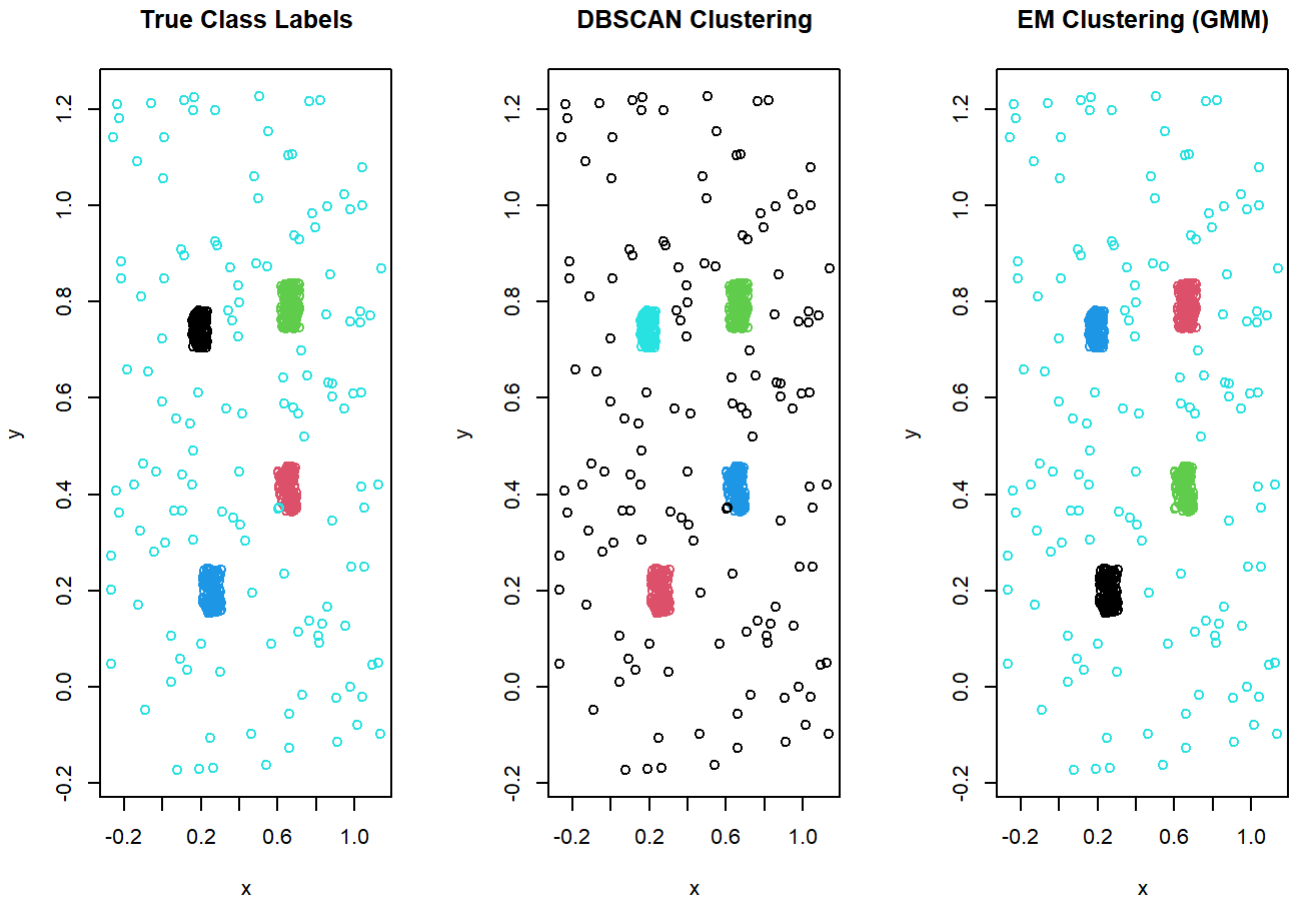
ได้ผลว่า DBSCAN และ EM สามารถแยกกลุ่มได้อย่างชัดเจน (100% และ 99.7% ตามลำดับ)

```
par(mfrow = c(1,3))

plot(dat2[,1:2], col = dat2$CLASS, main = "True Class Labels")

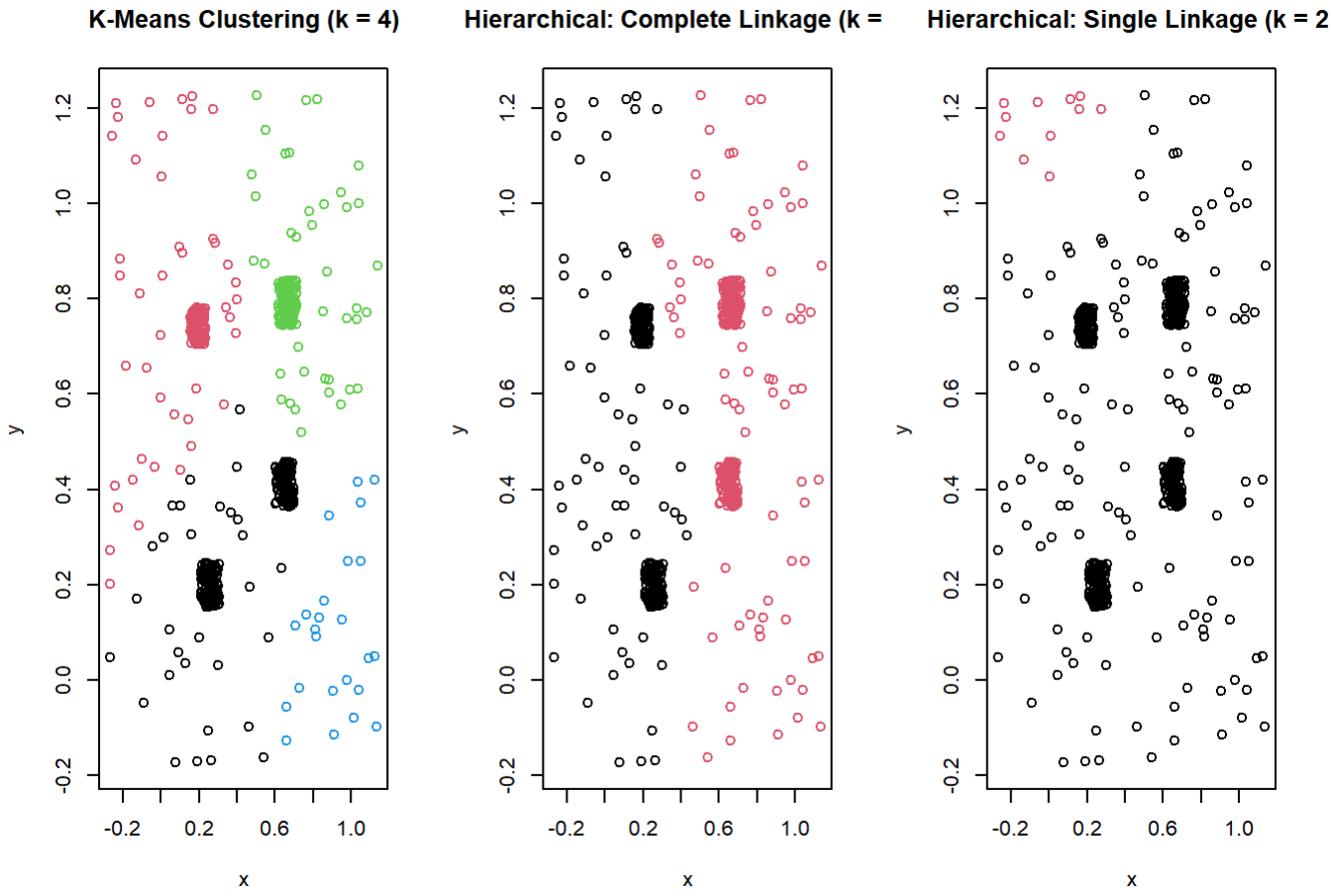
plot(dat2[,1:2], col = dat2_dbc$cluster + 1, main = "DBSCAN Clustering")

plot(dat2[,1:2], col = dat2_EM$classification, main = "EM Clustering (GMM)")
```



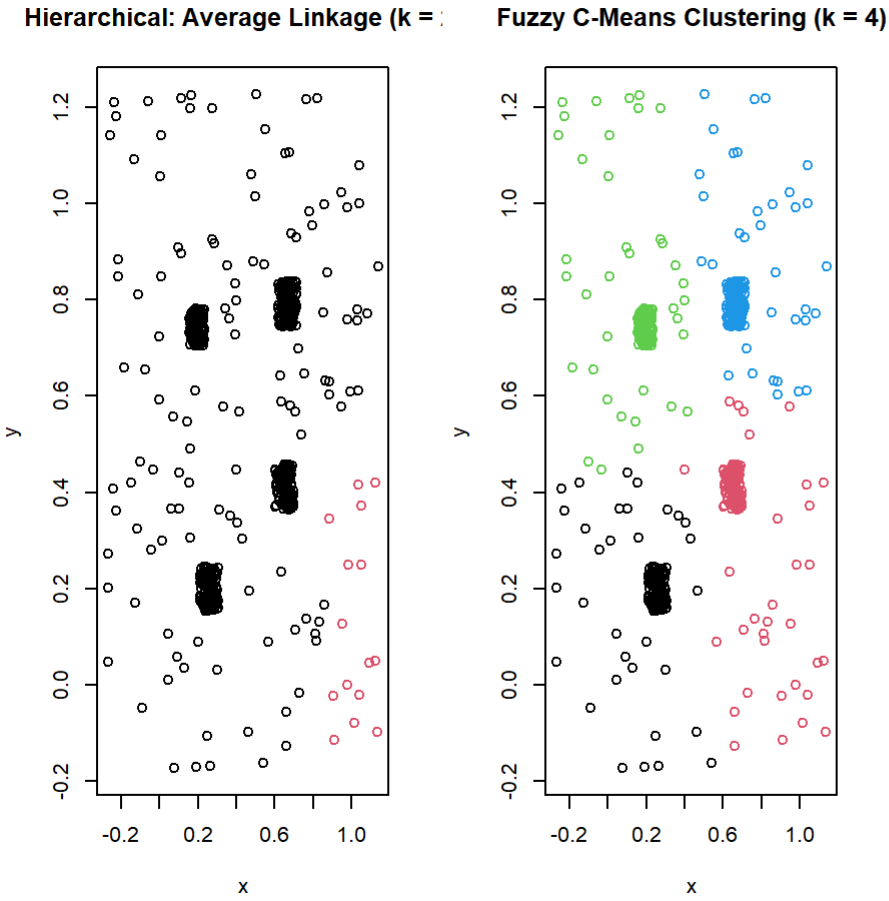

```
plot(dat2[,1:2], col = dat2_km$cluster, main = "K-Means Clustering (k = 4)")

plot(dat2[,1:2], col = cutree(dat2_com, 2), main = "Hierarchical: Complete Linkage (k = 2)")
plot(dat2[,1:2], col = cutree(dat2_avg, 2), main = "Hierarchical: Single Linkage (k = 2)")
```




```
plot(dat2[,1:2], col = cutree(dat2_sin, 2), main = "Hierarchical: Average Linkage (k = 2)")

plot(dat2[,1:2], col = dat2_fz$cluster, main = "Fuzzy C-Means Clustering (k = 4)")
```


Hierarchical: Average Linkage (k =
Fuzzy C-Means Clustering (k = 4)
y
x
-0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2
-0.2
0.2
0.6
1.0